TriZOD — Final Pipeline & Re-Referenced Dataset

Finalized pipeline · α-synuclein case study · chemical-shift perturbations

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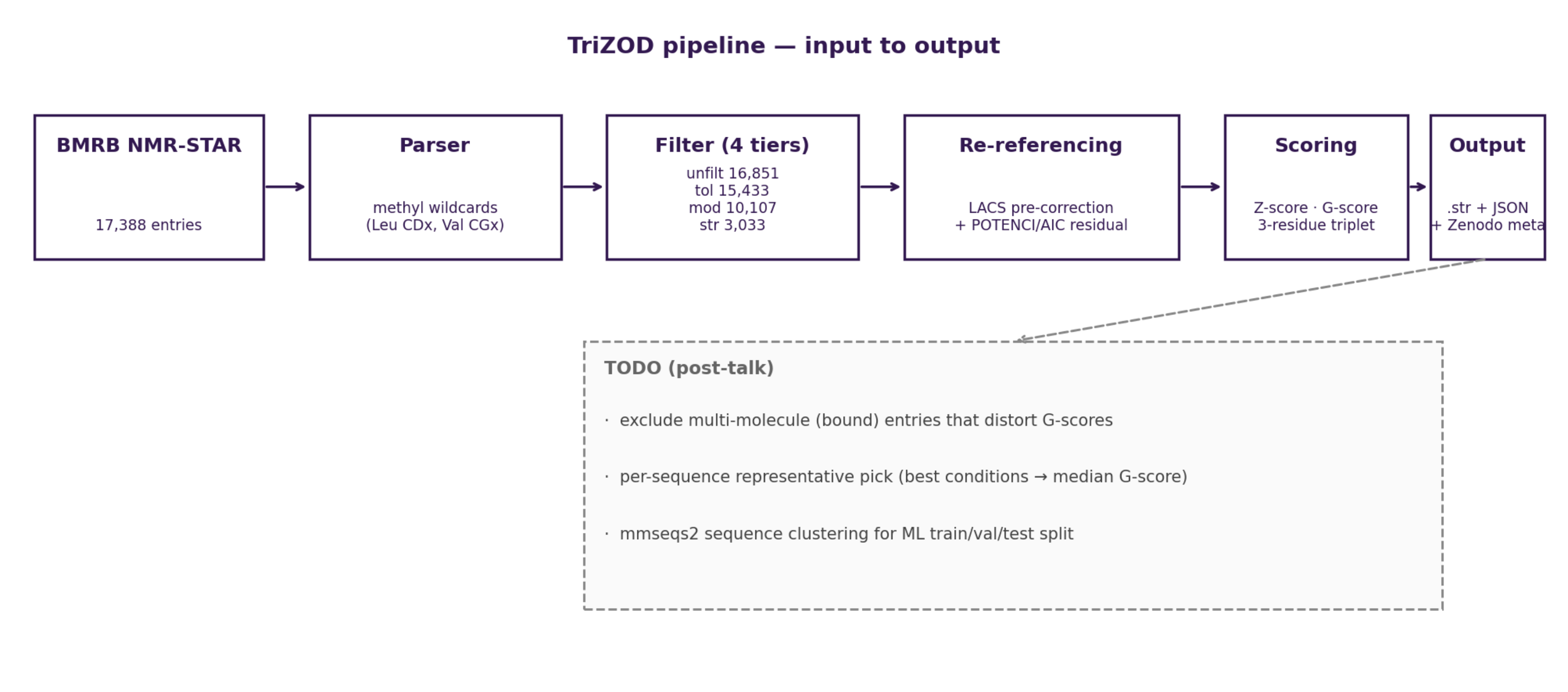
TUM · TriZOD project meeting

TriZOD pipeline at a glance

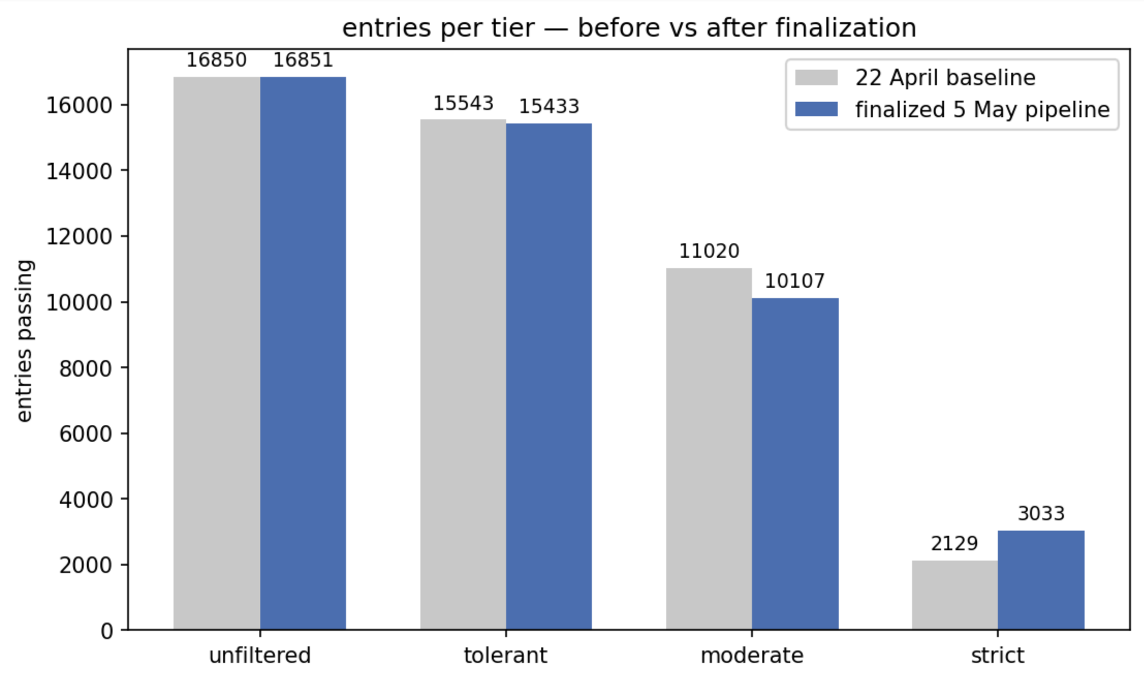


Goal: per-residue disorder scores (Z-score, G-score) from BMRB NMR shifts. Inputs at the left; output `.str` + JSON at the right; deferred work in the dashed branch.

What’s new since 22 April

- Methyl wildcards in the parser (Leu `CD1`/`CD2` ⟶ `CDx`, Val `CG1`/`CG2` ⟶ `CGx`)
- LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default `both`)
- New flag: `--emit-str <dir>` writes one re-referenced NMR-STAR per scored entity
- **Zenodo deposit metadata** in repo (DOI on first tagged release)

Filter improvements and per-tier dataset deltas



Removed denaturant false-positives, fixed the solid-state regex, added a paramagnetic-sample filter, relaxed `min-backbone-shift-types` 5⟶4 in strict, and broadened the Celsius heuristic.

Methyl wildcards: what and why

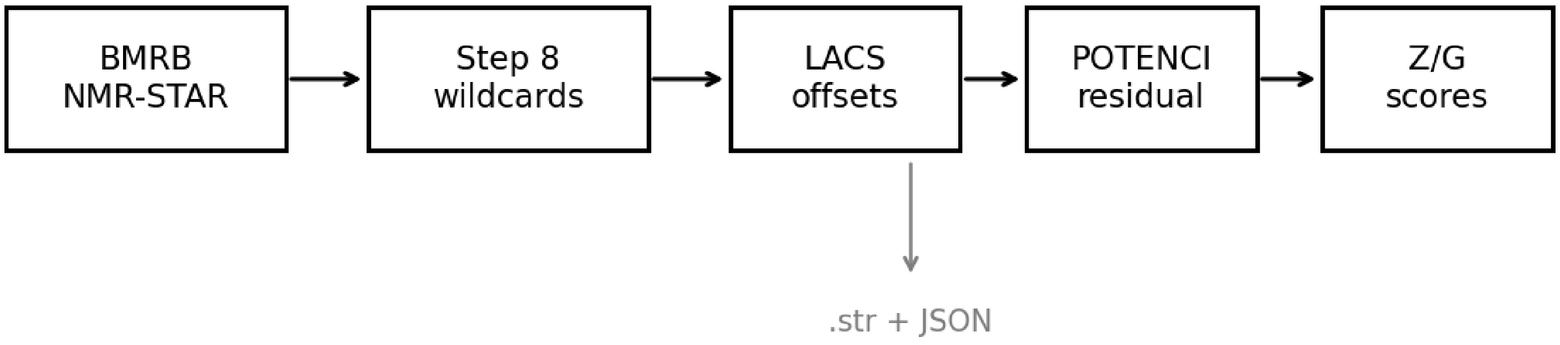
- Leucine: two methyls `CD1` + `CD2` (γ-carbon).
- Valine: two methyls `CG1` + `CG2` (β-carbon).
- **Geminal pairs are NMR-equivalent** — most pulse sequences cannot tell them apart.
- BMRB depositors must label one of each pair anyway: arbitrary stereospecificity.

Before: `LEU CD1 23.4 ppm` · `LEU CD2 24.1 ppm` (false claim of stereospecificity)

After (ambiguity ≠ 1): `LEU CDx 23.4 ppm` · `LEU CDx 24.1 ppm`

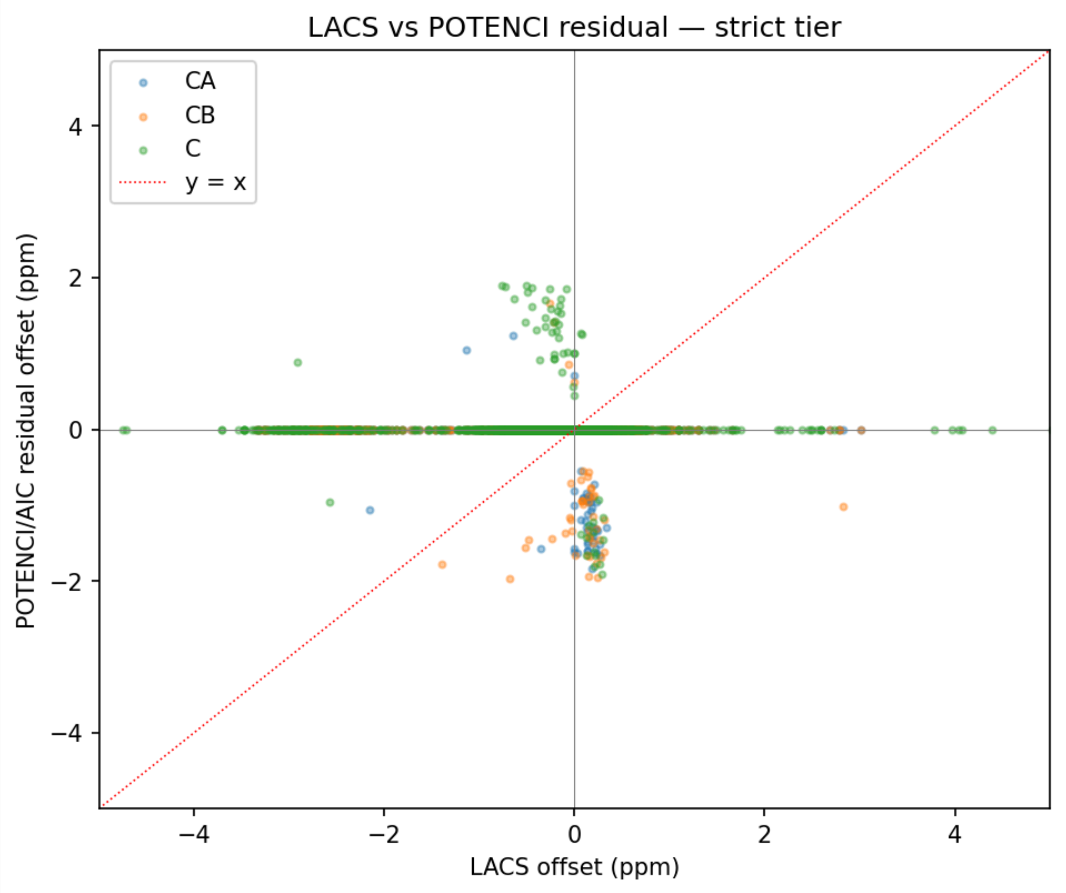
Backbone scoring is unchanged (these are side-chain methyls). The wildcards surface in the emitted `.str` files so downstream auto-assignment tools no longer propagate the false stereospecificity.

Re-referencing in the pipeline



raw shifts ⟶ **LACS pre-correction** (Wishart RC tables, robust line fits) ⟶ **POTENCI residual** (AIC-gated rolling 9-window) ⟶ Z / G-scores. Default mode is `both`.

LACS vs POTENCI residual capture

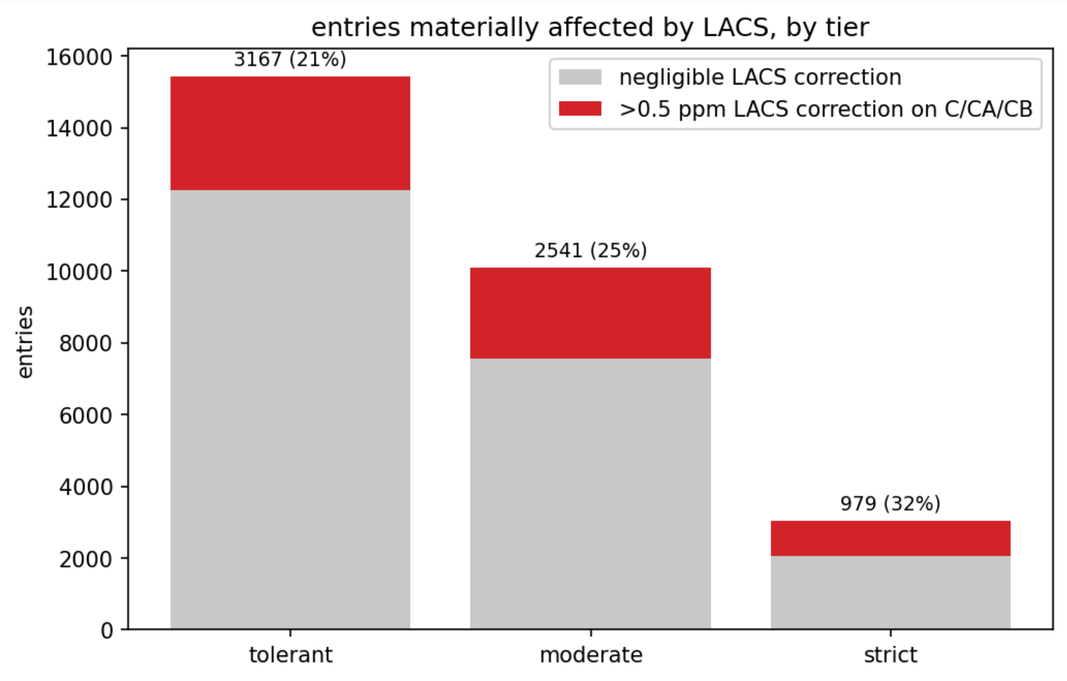


Each dot is **one entry × one atom** (CA / CB / C, strict tier). Not accumulated.

LACS catches the large systematic referencing bias.

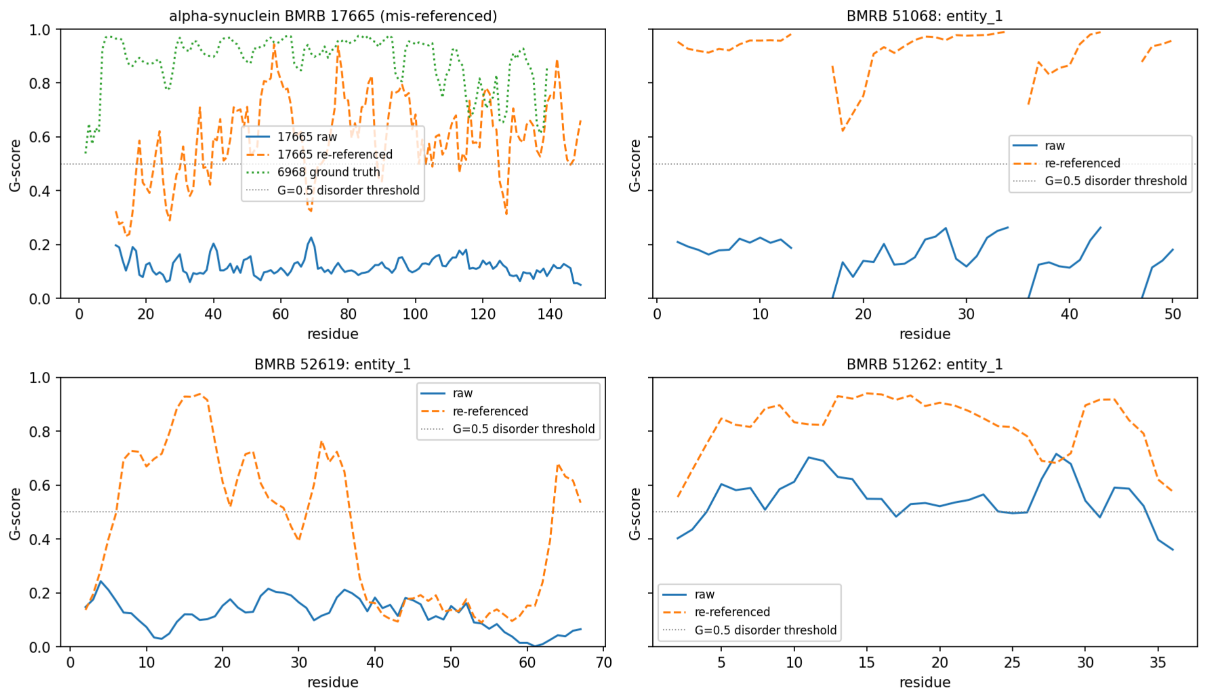
POTENCI/AIC residual mops up the remainder — most points land near the origin once LACS has done its job.

Entries materially affected by LACS



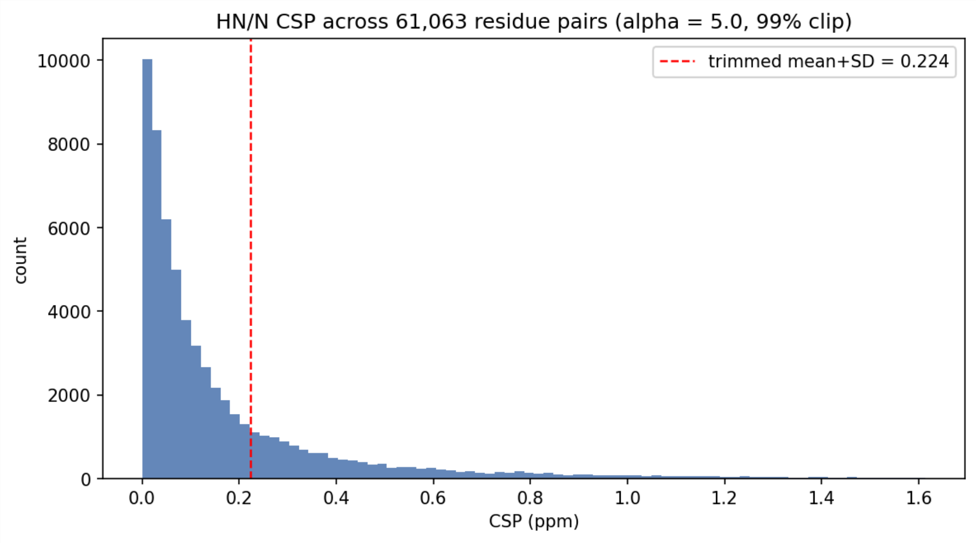
Red = entries with $|\text{LACS offset}| > 0.5$ ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for **21%** (tolerant), **25%** (moderate), **32%** (strict).

α-synuclein and top-3 G-score flippers

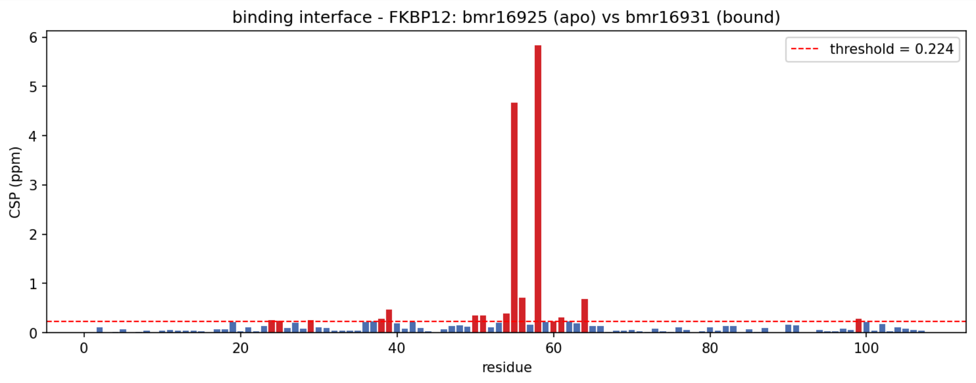


Panel A · BMRB 17665 (αSyn, mis-referenced) raw vs re-referenced + BMRB 6968 (αSyn ground truth). Panels B-D · top-3 flippers BMRB 51068 / 52619 / 51262. Gaps come from `k = 0` triplets (insufficient comparable shifts ⟶ G-score is NaN). αSyn LACS offset CA/CB = +2.82 ppm.

Reid #1: chemical-shift perturbations (residue-level)

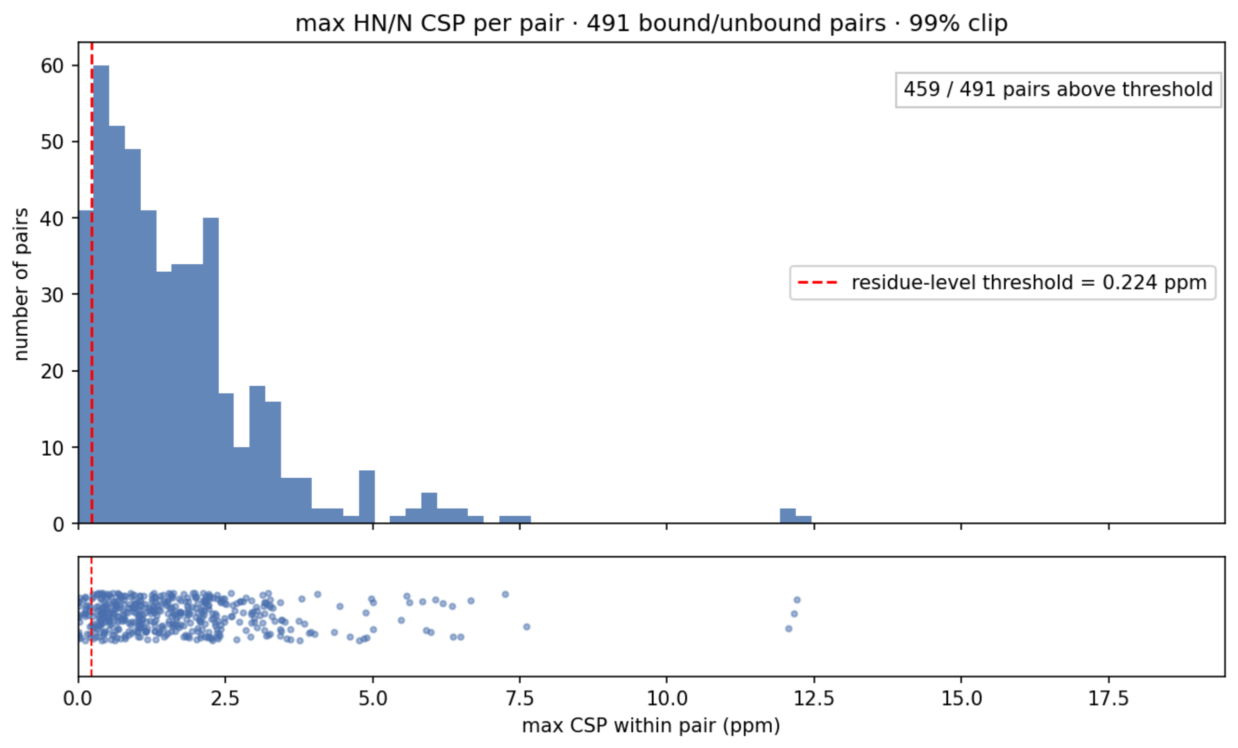


All 61,063 per-residue CSPs across 581 pairs. Threshold **0.224 ppm** = trimmed-mean+SD on the bottom 90% (top 10% dropped so the threshold isn't inflated by interface residues).



FKBP12 apo (bmr16925) vs bound (bmr16931). Two binding-pocket residues (55, 58) far above threshold.

Max CSP per pair · how strong is the strongest binding shift?



One value per pair (581 points): the **maximum** HN/N CSP within each pair. Pairs below the 0.224-ppm threshold are essentially silent; those above are real binding events.

What's still TODO

- **Multi-molecule entries distort G-scores** — exclude entries with a non-polymer or nucleic-acid binding partner from the dataset.
- **Per-sequence representative selection** — for the same protein with multiple BMRB entries, pick the one with best experimental conditions; tiebreak by median G-score.
- **mmseqs2 sequence clustering** for the ML train/val/test split (per the original CheZOD/TriZOD report).
- All three are flagged as TODO in the workflow on slide 2.

Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

Per-tier counts after the rerun:

Tier	Entries	Coverage
strict	3,033	17.00%
moderate	10,107	56.64%
tolerant	15,433	86.49%
unfiltered	16,851	94.45%

Next steps

- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate on Reid's CSP — extend to multi-atom $\Delta\omega_{\mathrm{RMS}}$ (Schumann/Williamson 2008)
- Reid follow-up with John Markley on why BMRB stopped LACS reports (frozen July 2020)
- Possible auto-assignment partner who'd benefit from `CDx`/`CGx` outputs